

A Heat-Related Illness Education Intervention for High School Athletes and Marching Band Members

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1. Introduction

Heat-related illnesses (HRIs), such as heat cramps, heat exhaustion, and heat stroke, are a critical health threat and can be fatal if not recognized and treated promptly (Casa and Stearns, 2017). They occur when heat exposure, physical exertion, or both exceed the body's ability to regulate temperature. Severe forms, especially heat stroke, can involve central nervous system dysfunction. In the U.S., at least 2,325 heat-related deaths were reported in 2023 (Howard et al., 2024), and more than 10,000 annual heat-related deaths were estimated for 1997–2006 (Weinberger et al., 2020).

HRI risk is amplified during sports and other vigorous activities in hot and humid conditions. HRIs are among the top three causes of athlete deaths (Lee-Chiong Jr. et al., 1995; Maron et al., 2009; Casa et al., 2012, 2015). From 2000 to 2024, at least 77 U.S. athletes died from HRIs, including 65 teenagers (84%) (Yancey-Bragg et al., 2024). More than 9,000 high school athletes are treated for HRIs each year (Yard et al., 2010). Marching band members also face HRI risk during prolonged outdoor rehearsals and performances, often while wearing full uniforms and carrying instruments (Kelly et al., 2025).

Insufficient heat acclimatization increases HRI risk among both athletes and marching band members (Kerr et al., 2013, 2019; Kelly et al., 2025). Therefore, HRI risk is not limited to hot climates or summer months; it can occur even before summer in cooler regions when students may not yet be fully acclimatized. HRI risk is shaped by environmental, individual, and activity-related factors. Environmental factors include high temperature and humidity, direct sun exposure, limited airflow, and radiant heat from sun-heated hard surfaces (e.g., asphalt and concrete). Individual and activity-related factors include dehydration, insufficient acclimatization, previous heat illness, poor sleep, intense or prolonged exertion, and heat-trapping uniforms and equipment (Casa et al., 2015; Garcia et al., 2022; Périard et al., 2022; Yamaguchi et al., 2024).

Since these factors interact, prevention requires more than hydration alone. It also depends on gradual heat acclimatization, environmental monitoring, adjustments in activity intensity and work-to-rest ratios, symptom recognition, and emergency action plans that specify rapid cooling before transport when heat stroke is suspected (Casa et al., 2015; CDC and NIOSH, 2026). The American Academy of Pediatrics also recommends regularly educating children and adolescents about proper preparation, hydration, honest symptom reporting, recovery, and rest (Bergeron et al., 2011).

This project proposes a community health education intervention focused on HRI prevention, recognition, and response among high school athletes and marching band members. The intervention will be delivered through a self-paced online educational module. HRI knowledge will be assessed among participants randomly assigned to either an intervention or

control group. Assessment items are aligned with six predefined, evidence-based learning objectives derived from published research and heat safety guidelines. Comparison of assessment scores between the two groups will estimate the intervention's short-term effect on participants' HRI knowledge. Follow-up assessments will examine whether knowledge gains are sustained over time and whether repeated exposure to the educational module can reinforce longer-term learning.

2. Background

Heat-related community interventions have taken diverse forms, including educational programs, environmental monitoring efforts, heat warning and communication initiatives, hydration and cooling practices, and outreach and social support programs. They have been implemented among diverse populations across a wide range of geographical settings (Johar et al., 2025; Buller et al., 2026; Hasan et al., 2021; Schapiro et al., 2024). Existing studies suggest that educational interventions generally improve heat-related health knowledge (Johar et al., 2025; Buller et al., 2026). However, evidence that these gains translate into sustained preventive behaviors or improved health outcomes is less conclusive. The methodological quality of these studies also remains limited (Buller et al., 2026). For example, few studies have used randomized controlled experiments. A notable exception is Razzak et al. (2022), who evaluate an educational intervention delivered by community health workers in low-income communities in Karachi, Pakistan, using a cluster-randomized controlled trial and report improvements in heat-related knowledge and preventive practices. The proposed study will similarly use a randomized controlled trial but focus specifically on high school students who face heat risk during exertional activities.

Most existing student-focused educational interventions address hydration practices rather than comprehensive heat risk education covering risk factors, symptom recognition, and emergency response (Schapiro et al., 2024; Kacem et al., 2026; Atkins et al., 2021). As a broader student-focused intervention, Li et al. (2022) evaluate an educational program on heatwaves and climate change in two elementary schools in Dongtai City, China, with one school assigned to intervention and the other to control, and report improvements in students' knowledge, attitudes, and preventive practices. In contrast, the proposed study will target high school students engaged in exertional activities and randomly assign individual students within participating schools to intervention and control groups, providing a stronger basis for causal comparison. To the best of the author's knowledge, no prior study in the US has delivered and evaluated a comprehensive educational intervention on HRI prevention, recognition, and response among high school athletes and marching band members.

3. Proposed Intervention and Evaluation

3.1 Study Design and Participants: This randomized controlled experiment will evaluate the effect of a self-paced educational module on HRI knowledge among high school athletes and marching band members. Participants in grades 10–12

will be recruited from the tennis teams, lacrosse teams, and marching band programs at Bedford High School and Milton High School in Massachusetts. Neither school offers preseason or early-season heat safety education for these activities, and such education appears limited across other varsity programs at both schools. Bedford varsity tennis players have experienced HRI symptoms in multiple recent seasons, including one case that prompted ambulance transport for evaluation.

Participants will be assigned to intervention and control groups in a 1:1 ratio using computer-generated randomization. Intervention-group participants will complete the knowledge assessment immediately after the module, while control-group participants will complete the same assessment before receiving access to the module. Participation will be voluntary and will not affect students' team or band membership, playing time, performance assignments, or standing. Information about grade level, activity group, and prior HRI education or experience will be collected to characterize the study population. No direct identifiers, such as names, email addresses, student IDs, or IP addresses, will be collected. Only aggregate findings will be reported, and individual responses will not be shared with supervising adults, such as coaches or band directors.

3.2 Educational Module: The educational intervention is a self-paced online module consisting of a 10-slide narrated presentation (10 minutes) and an accompanying handout. The module and assessment are structured around six learning objectives (Table 1) derived from published guidance on HRI prevention, recognition, and emergency response (Bergeron et al., 2011; Casa et al., 2015; Hewett Brumberg et al., 2024; CDC and NIOSH, 2026). The module explicitly distinguishes students' responsibilities from those of supervising adults and healthcare professionals. Students are not instructed to diagnose a specific HRI, establish heat safety policies, or independently manage practices and responses. Instead, the module emphasizes recognizing danger, stopping activity, reporting concerns, obtaining adult and emergency assistance, and assisting as directed. It also addresses social pressures that may encourage students to hide symptoms or continue participating.

3.3 Assessment and Procedure: HRI knowledge will be measured with an 18-item assessment aligned with the six learning objectives in Table 1. Three items assess each learning objective. Each item is a multiple-choice question. A correct response receives one point, while an incorrect or unanswered response receives zero points, producing a total score ranging from 0 to 18; higher scores indicate greater HRI knowledge. All participants will be instructed to complete the assessment without consulting the educational handout so that scores reflect unaided knowledge. After completing the assessment, control-group participants will also receive access to the module. This allows all participants to receive the educational materials.

3.4 Data Analysis: Assessment scores will first be summarized using descriptive statistics for the intervention and control groups, including mean and median scores, score range, standard deviation, and IQRs. Scores will also be summarized by learning objective to identify areas of stronger or weaker HRI knowledge. The intervention effect will then be estimated

using linear regression, with the total HRI knowledge score (0–18) as the dependent variable and intervention status as the explanatory variable, coded 1 for participants in the intervention group and 0 for participants in the control group. In this model, the coefficient for intervention status represents the estimated mean difference in assessment score attributable to the intervention. The corresponding *p*-value and 95% confidence interval will be reported to quantify statistical uncertainty. Longer-term learning retention will be evaluated through follow-up assessments that compare participants' scores over time. The effect of repeated exposure will be examined by comparing subsequent assessment scores between participants who receive additional exposure to the educational module and those who do not.

Table 1: Learning Objectives and Assessment Blueprint

Learning Objective (LO)	Operational description
LO1: Identify risk factors	Identify environmental, individual, and activity-related factors that increase HRI risk
LO 2: Select prevention strategies	Select appropriate actions before and during activity, including acclimatization, hydration, activity adjustment, rest, and clothing/equipment adjustments
LO 3: Recognize warning signs and stop activity	Recognize common symptoms and observable signs that require the affected student to stop participating and seek help
LO 4: Select reporting and peer-support actions	Report symptoms promptly, resist pressure to continue participating, support symptomatic peers, and remain with them until a supervising adult takes over
LO 5: Recognize possible heat stroke	Recognize altered mental status, loss of coordination, collapse, seizure, and other red flags as signs of a possible medical emergency
LO 6: Select the immediate emergency response	Stop activity, alert a supervising adult, activate the emergency plan and 911 response, begin rapid cooling as directed, and avoid unsafe fluid administration

3.5 Sustainability and Scalability: The intervention is designed for low-cost, repeated implementation because its educational module and assessment are digital and reusable. It requires only internet access and a web-enabled device, with no specialized equipment or paid instructional staff. The 10-minute module can be completed in a single session. Coaches, athletic directors, band directors, or school nurses can coordinate module distribution during preseason or early-season activities. If the initial intervention is successful, the materials can be expanded to additional teams, activities, and schools. Feedback from participating students and supervising adults, together with ongoing assessment results, will guide revisions while preserving the six evidence-based learning objectives. Maintaining the materials and implementation instructions in a shared online format will allow school personnel and future student leaders to continue the program beyond this project.

4. Preliminary Intervention Results and Feasibility

To provide preliminary evidence for the proposed intervention, an initial pilot was conducted with 20 high school athletes and marching band members, with 10 participants in each of the intervention and control groups. All participants completed

the intervention study. Because of the small sample, these results are interpreted as a preliminary estimate of the intervention's short-term effect and as an assessment of implementation feasibility.

The intervention group had a mean score of 17.3 (median: 18, standard deviation: 1.25, standard error: 0.40, 95% confidence interval: 16.41–18.19), compared with 12.2 in the control group (median: 12, SD: 1.75, SE: 0.55, 95% CI: 10.95–13.45). As shown in Fig. 1, intervention-group scores are consistently higher than control-group scores. Linear regression estimates an intervention effect of 5.1 points on the 18-point assessment, equivalent to 28.3 percentage points (SE: 0.68, 95% CI: 3.67–6.53, $p < 0.001$).

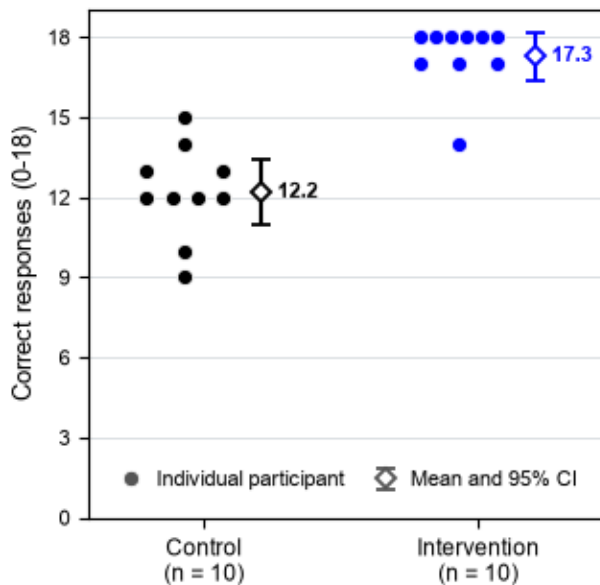


Fig. 1: Assessment Scores by Study Group

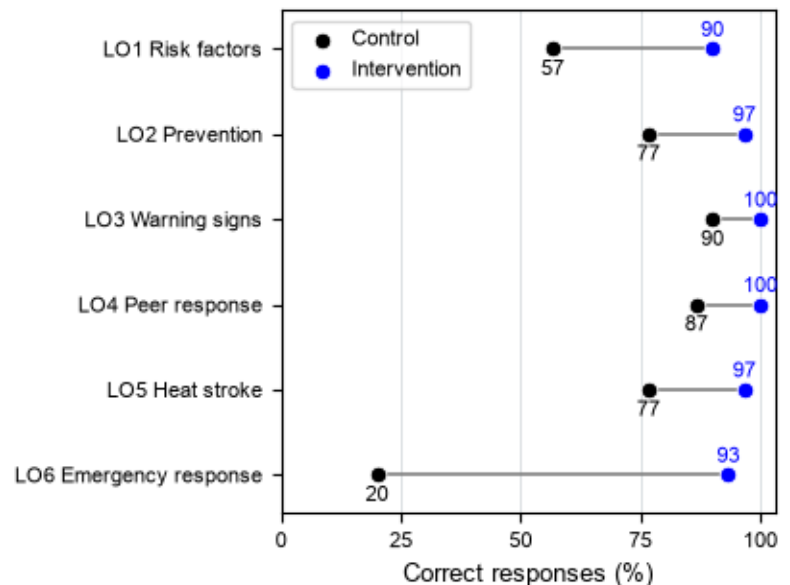


Fig. 2: Correct Responses by Learning Objective

Fig. 2 shows higher correct-response rates for the intervention group in all six learning objectives. Intervention-group accuracy ranges from 90% to 100%, compared with 20% to 90% in the control group, with the largest difference for emergency response (LO 6: 93% vs. 20%). Although preliminary, these patterns suggest that the intervention can improve HRI knowledge across multiple content areas, with a particularly substantial effect on emergency response knowledge.

5. Conclusion

HRIs pose a preventable but potentially life-threatening risk to high school athletes and marching band members. This project proposes an evidence-based, low-cost educational intervention designed to improve students' ability to prevent, recognize, and respond to HRIs. Preliminary results indicate a substantial short-term effect on HRI knowledge and demonstrate online implementation feasibility. With support from coaches, athletic and band directors, and other school personnel, the intervention can be incorporated into preseason or early-season heat safety education with minimal additional resources. Funding would support module revisions based on feedback from students and supervising adults, follow-up evaluation, and expansion to additional teams, activities, and schools; the reusable format would keep recurring delivery costs minimal.

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